

LIE SYMMETRIES OF NONLINEAR PARTIAL DIFFERENTIAL EQUATIONS FOR OPTICAL SOLITONS

AKITO KOIKE¹, MANI SHAYESTEHFAR¹

¹ *The University of Sydney, School of Physics A28, Sydney 2006, Australia*

² *Department of Mathematical and Physical Sciences, La Trobe University, Bendigo 3552, Victoria, Australia*

³ *College of Science and Engineering, Flinders University, Flinders at Tonsley, South Australia 5042, Australia*

ABSTRACT. The Nonlinear Schrödinger Equation (NLSE) can be used to model the propagation of optical solitons in nonlinear fibers. The NLSE can be modified by considering higher order dispersion effects. The solutions to some variations of the NLSE have not previously been investigated. In this paper, we use Lie symmetries to find all of the solutions to a given differential equation at the same time. We explicitly utilise it for a pure, quartic soliton, then generalise to any higher order soliton. We find that in both cases, there are precisely two symmetries, both of which have already been found in experiment. This shows that no other solutions exist for those variations of the NLSE.

INTRODUCTION

The one-dimensional Nonlinear Schrödinger Equation (NLSE) is a well-established integrable second-order classical field equation [5] that models various physical phenomena, including Bose-Einstein condensates[3], Langmuir waves [7], and, most notably for our focus, the propagation of optical solitons in nonlinear fibers [1]. The study of the Nonlinear Schrödinger Equation has been of interest due to its ability to model the interplay between dispersion and Kerr nonlinearity, which governs wave propagation in various optical systems. Recent advancements have extended the NLSE by considering higher-order dispersion effects, leading to the discovery of quartic solitons with fourth-order dispersion [2, 9], such as those seen in Figure 1, as well as solitons up to the tenth-order which have been experimentally realised [10]. Such variations of the NLSE have led to new frontiers in ultra-fast laser technologies and optical communication systems and promote innovations such as in high-speed data transmission [6]. Subsequently, given one solution to variations of the NLSE, it is certainly of interest to investigate whether further solutions can be generated, and if so, what properties do these solutions exhibit.

In this study we focus on understanding the solutions to the generalisation of high-order dispersion solitons using the novel method of Lie Symmetries, by exploring variations of the NLSE. This approach relies on the concept that if a differential equation possesses symmetry, its solutions can be transformed while maintaining that symmetry. In other words, knowing the symmetries of a differential equation allows us to simplify it through transformations, making it easier to solve. Further understanding the solutions to families of optical solitons in particular could have profound implications including enhancements in optical communication systems. We first introduce the

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physical background of the optical soliton in section 1. In section 2 we introduce the generalised form of the NLSE and the particular constraint of interest. In section 3 we introduce Lie symmetries its necessary blueprint for ODEs. Section 4.1 involves finding the symmetries of the Pure Quartic Soliton (PQS) and section 4.2 includes our generalisation of the pure-form soliton.

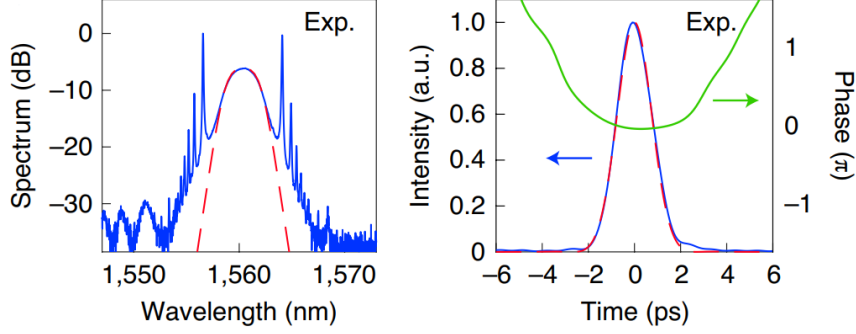


FIGURE 1. Measured fourth-order dispersion solitons from [9]. **Left:** Output spectra (solid blue) with corresponding soliton and PQS spectral shape (dashed red). **Right:** Retrieved (solid blue) and theoretical (dashed red) temporal intensity profiles, with the recovered temporal phases (green line).

1. OPTICAL SOLITONS

An optical Soliton is a stable localised wave packet that remains undistorted over long distances due to balancing effects of nonlinearity and negative even dispersion [4]. Nonlinearity occurs due to the Kerr effect in which the refractive index of a material changes with the intensity of the light passing through it, as

$$n(I) = n_0 + n_2 I \quad (1)$$

where n is the refractive index, I is the intensity of the light field, and n_0, n_2 are linear and nonlinear refractive index constants. The phase velocity of light, given by $v = c/n$ (c is the speed of light in vacuum) is inversely related to refractive index and thus

$$v(I) \propto \left(1 + \frac{n_2}{n_0} I\right)^{-1}. \quad (2)$$

Hence, the high-intensity peak of a pulse will have the lowest phase velocity, while the low-intensity edge will have the higher phase velocity. This difference in velocity in the pulse will lead to blue-shifting around the trailing edges of the wave packet, while the leading edge is red-shifted.

Dispersion of light is due to the dependence of the refractive index in a medium to frequency. In particular, the propagation constant of light $\beta(\omega)$ is

$$\beta(\omega) = \frac{n(\omega)\omega}{c}. \quad (3)$$

Generally, optical solitons are formed in regimes where the frequency spectrum has a limited width, with each pulse containing hundreds of optical periods with fractional bandwidths $\Delta\omega/\omega < 10^{-2}$. With a small bandwidth approximation $\beta(\omega)$ can be Taylor expanded around a central frequency $\omega_0 \approx 1.2 \times 10^{15} \text{ s}^{-1}$ [4] as

$$\beta(\omega) = \beta(\omega_0) + \beta_1(\omega - \omega_0) + \frac{\beta_2}{2!}(\omega - \omega_0)^2 + \frac{\beta_3}{3!}(\omega - \omega_0)^3 + \dots \quad (4)$$

Since the group velocity is $1/v_g = \partial\beta/\partial\omega$, then

$$\frac{1}{v_g} = \beta_1(\omega - \omega_0) + \frac{\beta_2}{2!}(\omega - \omega_0)^2 + \frac{\beta_3}{3!}(\omega - \omega_0)^3 + \dots \quad (5)$$

In order to form a stable optical soliton, the red-shifted leading edge must slow down and the blue-shifted trailing edge must be sped up. This is precisely due to the effects of dispersion and only occurs if Eq. (5) is monotonically decreasing in ω . Therefore the condition forced on β is such that all odd β coefficients must vanish and all even β coefficients should be negative or zero [4].

2. THE GENERALISED NLSE & STATIONARY REDUCTION

Recalling the famous one-dimensional Linear Schrödinger Equation (LSE),

$$i\hbar \frac{\partial\psi}{\partial t} + \frac{\hbar^2}{2m} \frac{\partial^2\psi}{\partial x^2} - V\psi = 0, \quad (6)$$

the NLSE is analogous to this equation, with the difference that the kinetic energy term is replaced with dispersion, and the potential energy is nonlinear and so is dependent on ψ since the optical soliton traps itself in its own potential well. The NLSE is written as

$$i \frac{\partial\psi}{\partial z} - \frac{\beta_2}{2} \frac{\partial^2\psi}{\partial T^2} + \gamma|\psi|^2\psi = 0, \quad (7)$$

where $\psi(z, T)$ is the soliton wave envelope, z is the axis of propagation, β_2 is the dispersion coefficient, $\gamma > 0$ is the nonlinear coefficient, and T^1 is co-moving time. Since the choice of z, t are mostly arbitrary, one can exchange these parameters in NLSE compared to LSE for experimental purposes. This is mainly because in Eq. (7), z is taken to be the evolution parameter to reflect the way experiments are carried out.

The NLSE only includes the effects of β_2 , but the differential equation can be further generalised into the Generalised Nonlinear Schrödinger Equation (GNLSE) by allowing the addition of higher-order even dispersion terms up to order n ,

$$i \frac{\partial\psi}{\partial z} + \sum_{k=2,4,\dots}^n (-1)^{\frac{k}{2}} \frac{\beta_k}{k!} \frac{\partial^k\psi}{\partial T^k} + \gamma|\psi|^2\psi = 0. \quad (8)$$

In the stationary reduction of the GNLSE, the ansatz $\psi(z, t) = u(t)e^{i\mu z}$ (note the $T \rightarrow t$ substitution) is used so that the soliton shape is preserved with propagation and μ is the nonlinear phase shift parameter, used to identify members of the family of solutions [11]. Substituting this ansatz into Eq. (8) yields

$$-\mu u + \sum_{k=2,4,\dots}^n (-1)^{\frac{k}{2}} \frac{\beta_k}{k!} \frac{\partial^k u}{\partial t^k} + \gamma u^3 = 0. \quad (9)$$

This reduction conveniently reduces a partial differential equation (PDE) to a nonlinear ordinary differential equation (ODE) for which there are various known methods of solution. For example, analytic methods have been discovered that find stable solitons, as well as higher-order solutions that tend to be unstable and that are difficult to find numerically. However, given a solution, such methods cannot provide any insight about the uniqueness of the provided solution, or whether “symmetries” in the solution exists (i.e. whether changes in a variable can preserve the solution).

¹For simplicity, we replace T which describes the profile of the soliton wave, with time t by a change of variable.

3. LIE SYMMETRIES METHOD

Lie symmetries provide a rigorous method to finding general solutions to differential equations by utilising the symmetries inherent in these equations. Developed by Sophus Lie, the method is based on the idea that differential equations remain invariant under certain continuous transformations of the independent and dependent variables. Mathematically, these transformations form a local Lie group of transformations, and their infinitesimal generators take the form of vector fields [8]. The following provides the groundwork required to apply this method on GNLSEs. Detailed description of definitions is provided in Appendix A.

Consider an n -th order ODE, $F(x, y, y', \dots, y^{(n)}) = 0$ where $y^{(k)} = \frac{d^k y}{dx^k}$ and $k = 1, 2, \dots, n$. The one-parameter Lie Group of point transformations is

$$\tilde{x} = x + \varepsilon \xi(x, y) + O(\varepsilon^2) \quad (10a)$$

$$\tilde{y} = y + \varepsilon \phi(x, y) + O(\varepsilon^2) \quad (10b)$$

with the group parameter ε and a corresponding infinitesimal generator

$$X = \xi(x, y) \frac{\partial}{\partial x} + \phi(x, y) \frac{\partial}{\partial y}. \quad (11)$$

The infinitesimal generator is a vector field that encapsulates the “infinitesimal” behaviour of the group action near the identity element. The Lie group in Eqs. (10) is only admitted by the ODE if and only if the invariance condition below holds:

$$X^{(n)} (F(x, y, y', \dots, y^{(n)})) = 0 \quad \text{whenever} \quad F(x, y, y', \dots, y^{(n)}) = 0. \quad (12)$$

Here $X^{(n)}$ is the n -th prolongation of the infinitesimal generator which extends X to act on the derivatives of y by treating them as independent variables up to the n -th order in the ODE. In this case the prolongation is given by

$$X^{(n)} = X + \phi^{(1)}(x, y, y') \frac{\partial}{\partial y'} + \dots + \phi^{(n)}(x, y, y', \dots, y^{(n)}) \frac{\partial}{\partial y^{(n)}} \quad (13)$$

where recursively we have

$$\phi^{(k)}(x, y, y', \dots, y^{(k)}) = \frac{d}{dx} (\phi(x, y) - \xi(x, y) y') + \xi \frac{\partial y^{(k)}}{\partial x} \quad (14)$$

with $\phi^{(0)} = \phi(x, y)$, and $\frac{d}{dx}$ is the k -th order total derivative with respect to x , defined by

$$\frac{d}{dx} = \frac{\partial}{\partial x} + y' \frac{\partial}{\partial y} + y'' \frac{\partial}{\partial y'} + \dots + y^{(k+1)} \frac{\partial}{\partial y^{(k)}} \quad (15)$$

4. LIE SYMMETRIES OF PURE-FORM NLSE

Given the recipe to construct Lie symmetries of ODEs, we now implement this method on a variation of the ODE from Eq. (9). This variation is the ‘pure’ form of the n -th order NLSE where $\beta_i = 0$ if $i \neq n$. We present a particular example before moving on to a generalisation proposition.

4.1. Pure Quartic Soliton (PQS). For the pure quartic soliton (fourth order NLSE), the ODE is simplified to

$$-\mu u + \tilde{\beta}_4 u_{tttt} + \gamma u^3 = 0 \quad (16)$$

where $\tilde{\beta}_4 := \beta_4/4! < 0$, and the notation for derivatives is adjusted for readability. Recall that the stationary reduction ansatz resulted in the independence of the ODE from the z variable. Hence in a

particular constraint where the lack of $z \implies$ lack of μ , we shall set $\mu = 0$. In this case Eq. (16) simplifies to

$$F(t, u) := u_{tttt} + \frac{\gamma}{\beta_4} u^3 = 0. \quad (17)$$

The one-parameter Lie group of point transformations is:

$$\tilde{t} = t + \varepsilon \xi(t, u) + O(\varepsilon^2) \quad (18a)$$

$$\tilde{u} = u + \varepsilon \phi(t, u) + O(\varepsilon^2) \quad (18b)$$

with the infinitesimal generator,

$$X = \tau(t, u) \frac{\partial}{\partial t} + \phi(t, u) \frac{\partial}{\partial u}. \quad (19)$$

Following the method, since F is a fourth order ODE, then the fourth order prolongation is

$$\begin{aligned} X^{(4)} = X &+ \phi^t(t, u, u_t) \frac{\partial}{\partial u_t} + \phi^{tt}(t, u, u_t, u_{tt}) \frac{\partial}{\partial u_{tt}} \\ &+ \phi^{ttt}(t, u, u_t, u_{tt}, u_{ttt}) \frac{\partial}{\partial u_{ttt}} + \phi^{tttt}(t, u, u_t, u_{tt}, u_{ttt}, u_{tttt}) \frac{\partial}{\partial u_{tttt}}. \end{aligned} \quad (20)$$

Applying this prolongation on F yields the important constraint

$$\phi^{(4)} + \frac{3\gamma}{\tilde{\beta}_4} \phi u^2 = 0 \quad (21)$$

At this point, using the formula from Eq. (14) ϕ^{tttt} can be calculated as

$$\phi^{tttt} = \frac{d^4}{dt^4} (\phi(t, u) - \tau(t, u) u_t) + \tau(t, u) \frac{\partial u_{tttt}}{\partial t} \quad (22)$$

$$\begin{aligned} &= u_t (-6u_{tt} (3\tau_{ttu} - 2\phi_{ttu}) - 15u_{tt}^2 \tau_{uu} - 4u_{ttt} (4\tau_{tu} - \phi_{uu}) - \tau_{tttt} - 5u_{tttt} \tau_u + 4\phi_{tttu}) \\ &+ u_t^2 (-6u_{tt} (4\tau_{tuu} - \phi_{uuu}) - 10u_{ttt} \tau_{uu} - 2(2\tau_{tttu} - 3\phi_{tttu})) \\ &+ u_t^3 (-10u_{tt} \tau_{uuu} - 2(3\tau_{ttuu} - 2\phi_{tuu})) + u_{tttt} (\phi_u - 4\tau_t) + u_t^4 (\phi_{uuuu} - 4\tau_{tuu}) \\ &+ u_t^5 (-\tau_{uuuu}) + u_{tt} (-2(2\tau_{ttt} - 3\phi_{ttu}) - 10u_{ttt} \tau_u) - 2u_{ttt} (3\tau_{tt} - 2\phi_{tu}) \\ &- 3u_{tt}^2 (4\tau_{tu} - \phi_{uu}) + \phi_{tttt}. \end{aligned} \quad (23)$$

Now substituting the constraint from Eq. (21) and separating each monomial's coefficient constructs the table

Label	Monomial	Coefficient
(a)	Constants	$\phi_{tttt} = 3\frac{\gamma}{\beta_4}\phi u^2 + 4\frac{\gamma}{\beta_4}u^3\tau_t - \frac{\gamma}{\beta_4}u^3\phi_u$
(b)	u_t	$-\tau_{tttt} + 4\phi_{tttu} = 5\gamma u^3\tau_u$
(c)	u_t^2	$4\tau_{tttu} = 6\phi_{ttuu}$
(d)	u_t^3	$6\tau_{ttuu} = 4\phi_{tuuu}$
(e)	u_t^4	$\phi_{uuuu} = 4\tau_{uuuu}$
(f)	u_t^5	$\tau_{uuuu} = 0$
(g)	u_{tt}	$4\tau_{ttt} = 6\phi_{ttu}$
(h)	u_{tt}^2	$12\tau_{tu} = 3\phi_{uu}$
(i)	u_{ttt}	$6\tau_{tt} = 4\phi_{tu}$
(j)	$u_t u_{tt}$	$12\phi_{tuu} = 18\tau_{ttu}$
(k)	$u_t u_{ttt}$	$4\phi_{uu} = 16\tau_{tu}$
(l)	$u_t u_{tt}^2$	$15\tau_{uu} = 0$
(m)	$u_t^2 u_{tt}$	$6\phi_{uuu} = 24\tau_{tuu}$
(n)	$u_t^3 u_{tt}$	$10\tau_{uuu} = 0$
(o)	$u_t^2 u_{ttt}$	$10\tau_{uu} = 0$
(p)	$u_{tt} u_{ttt}$	$10\tau_u = 0$

TABLE 1. Table of coefficients for a PQS with only β_4 term.

which is a set of 16 partial differential equations that can be solved by hand or a mathematical computational program such as Maple or Mathematica. The solution yields

$$\tau(t) = a_1 t + a_0 \quad (24)$$

$$\phi(u) = -2a_1 u \quad (25)$$

for arbitrary constants a_0, a_1 . This yields $X = (a_1 t + a_0)\partial_t + (-2a_1 u)\partial_u$, which is spanned by

$$X_1 = \partial_t \quad (26)$$

$$X_2 = t\partial_t - 2u\partial_u \quad (27)$$

By exponentiation of vector fields (which can be done by hand or by Maple), the one-parameter groups generated by each vector field are

$$G_1 : (z, t + \varepsilon, u) \quad (28)$$

$$G_2 : (z, e^\varepsilon t, e^{-2\varepsilon} u) \quad (29)$$

Finally, given $u = f(z, t)$ is a solution to the original differential equation, so are the functions

$$u^{(1)} = f(z, t - \varepsilon) \quad (30)$$

$$u^{(2)} = e^{2\varepsilon} f(z, e^{-\varepsilon} t) \quad (31)$$

Furthermore, an exact, group-invariant solution (see [8] for details) can be found.

Let $X = X_1 + kX_2 = (t + k)\partial_t + (-2u)\partial_u$ for some constant k that preserves the relative ratio of X_1 and X_2 . The characteristic equations yield

$$\frac{dt}{t + k} = \frac{du}{-2u} \quad (32)$$

Solving the equation above gives

$$u(t) = \frac{A}{(t + k)^2} \quad (33)$$

where A is a constant. Thus, $u(t) = \frac{A}{(t+k)^2}$ is a solution to the PQS in the stationary reduction case particularly when $\mu = 0$.

4.2. Generalised Pure-Form Soliton. In general, the even order pure soliton ODE is simplified to

$$-\mu u + \tilde{\beta}_n u_{t^{(n)}} + \gamma u^3 = 0 \quad (34)$$

where n is a positive, even integer, and $\tilde{\beta}_n := \beta_n/n! < 0$, and $t^{(n)}$ denotes n partial derivatives with respect to t . Similarly to the quartic soliton, set $\mu = 0$. In this case Eq. (34) simplifies to

$$F(t, u) := u_{t^{(n)}} + \alpha u^3 = 0. \quad (35)$$

where $\alpha := \frac{\gamma}{\tilde{\beta}_n}$. Again, the infinitesimal generator is

$$X = \tau(t, u) \frac{\partial}{\partial t} + \phi(t, u) \frac{\partial}{\partial u}. \quad (36)$$

Following the method, since F is an n th order ODE, then the n th order prolongation is

$$X^{(n)} = X + \phi^t \frac{\partial}{\partial u_t} + \phi^{tt} \frac{\partial}{\partial u_{tt}} + \dots + \phi^{t^{(n)}} \frac{\partial}{\partial u_{t^{(n)}}}. \quad (37)$$

Applying this prolongation on F yields the important constraint

$$\phi^{t^{(n)}} + 3\alpha\phi u^2 = 0 \quad (38)$$

At this point, using the formula from Eq. (14) $\phi^{t^{(n)}}$ can be calculated as

$$\phi^{t^{(n)}} = \frac{d^n}{dt^n} (\phi(t, u) - \tau(t, u) u_t) + \tau(t, u) \frac{\partial u_{t^{(n)}}}{\partial t} \quad (39)$$

Proposition. We claim that substituting the constraint from Eq. (38) and separating each monomial's coefficient always yields the following equations (for $n \geq 3$. However apart from the table, it turns out that the $n = 2$ case also matches the results obtained below.)

Label	Monomial	Coefficient
(a)	Constants	$\phi_{t^{(n)}} + \alpha\phi_u u^3 - 3\alpha\phi u^2 = n\alpha\tau_t u^3$
(b)	$u_{t^{(n-2)}}$	$\frac{n(n-1)}{2} \phi_t t u = \frac{(n-2)(n-1)n}{6} \tau_{ttt}$
(c)	$u_{t^{(n-1)}}$	$n\phi_t u = \frac{n(n-1)}{2} \tau_{tt}$
(d)	$u_t u_{t^{(n-1)}}$	$k_2 \phi_u u = \tau_u$
(e)	$u_{tt} u_{t^{(n-1)}}$	$0 = -k_1 \tau_u$

TABLE 2. Part of the table of coefficients for a PQS with only β_n term.

where $k_1, k_2 \in \mathbb{R}^+$. Solving by hand or by computer software yields

$$\tau(t) = a_1 t + a_0 \quad (40)$$

$$\phi(u) = -\frac{n}{2} a_1 u \quad (41)$$

for arbitrary constants a_0, a_1 . This yields $X = (a_1 t + a_0) \partial_t + (-\frac{n}{2} a_1 u) \partial_u$, which is spanned by

$$X_1 = \partial_t \quad (42)$$

$$X_2 = t \partial_t - \frac{n}{2} u \partial_u \quad (43)$$

By exponentiation of vector fields, given $u = f(z, t)$ is a solution to the original differential equation, so are the functions

$$u^{(1)} = f(z, t - \varepsilon) \quad (44)$$

$$u^{(2)} = e^{\frac{n}{2}\varepsilon} f(z, e^{-\varepsilon}t) \quad (45)$$

Furthermore, an exact, group-invariant solution (see [8] for details) can be found.

Let $X = X_1 + kX_2 = (t + k)\partial_t + (-\frac{n}{2}u)\partial_u$ for some constant k that preserves the relative ratio of X_1 and X_2 . The characteristic equations yield

$$\frac{dt}{t + k} = \frac{du}{-\frac{n}{2}u} \quad (46)$$

Solving the equation above gives

$$u(t) = \frac{A}{(t + k)^{\frac{n}{2}}} \quad (47)$$

where A is a constant. Thus, $u(t) = \frac{A}{(t+k)^{\frac{n}{2}}}$ is a solution to the general, pure-form soliton in the stationary reduction case particularly when $\mu = 0$.

Proof. Consider some term r in the expansion of this Eq. (39) grouped in like monomials. Note that r is a function of t, u and its partial derivatives with respect to t , and τ, ϕ and their partials with respect to both u and t (the wording u and its partial derivatives with respect to t refers to $u, u_t, u_{tt}, \dots$). Then (if the monomial term of r is not constant), then r is of the form

$$r = f \cdot m = f \prod_{i=1}^p u_{t^{(k_i)}} \quad (48)$$

where $k_i \in \mathbb{Z}^+$, m is the monomial term, and f is a function of u , and τ, ϕ and their partials with respect to both u and t . Define the order of the term r as

$$Q(r) := \sum_{i=1}^p k_i \quad (49)$$

and the width of the term r as

$$W(r) := p \quad (50)$$

For example, the monomial $u_t u_{t(2)}^3 u_{t(5)}$ has order $1 + 2 \cdot 3 + 5 = 12$, and width $1 + 3 + 1 = 5$. If the monomial term of r is constant, then $m = 1$, and $Q(r) = 0 = W(r)$.

Now for non-negative integers k and j , consider the function

$$g(j, k) = c \phi_{t^{(j)}u^{(k)}} \quad (51)$$

for some positive integer c . Let h be a monomial term, so

$$h(d_1, d_2, \dots, d_q) = \prod_{i=1}^q u_{t^{(d_i)}} \quad (52)$$

where $d_i \in \mathbb{Z}^+$ (Again if h is chosen to be the constant monomial term, then $h = 1$). Now consider the term $s_0 = g \cdot h$. Take the total derivative with respect to t , using product rule,

$$\frac{ds_0}{dt} = \frac{d}{dt}(g \cdot h) = h \frac{dg}{dt} + g \frac{dh}{dt} \quad (53)$$

Now

$$h \frac{dg}{dt} = h(g_t + g_u u_t) = h(\phi_{t^{(j+1)}u^{(k)}} + \phi_{t^{(j)}u^{(k+1)}} u_t) = (\phi_{t^{(j+1)}u^{(k)}})h + (\phi_{t^{(j)}u^{(k+1)}})h u_t \quad (54)$$

Split into like monomials, and let $s_1 = (\phi_{t^{(j+1)}u^{(k)}})h$, and $s_2 = (\phi_{t^{(j)}u^{(k+1)}})hu_t$.

Now by the product rule for multiple product of functions,

$$g \frac{dh}{dt} = g \frac{d}{dt} \left(\prod_{j=1}^q u_{t^{(d_j)}} \right) = \sum_{i=1}^q \left(g \prod_{j=1}^q u_{t^{(d_j + \delta_{ij})}} \right) \quad (55)$$

where δ_{ij} is the Kronecker delta. For $1 \leq i \leq q$, let $s_{q_i} = g \prod_{j=1}^q u_{t^{(d_j + \delta_{ij})}}$, so $\sum_{i=1}^q s_{q_i} = g \frac{dh}{dt}$

Notice that s_i all have the same form, that is

$$s_i = g(j_i, k_i)h(d_{i_1}, d_{i_2}, \dots, d_{i_\ell}) \quad (56)$$

for some non-negative integer ℓ . So $\frac{ds_0}{dt} = s_1 + s_2 + \sum_{i=1}^q s_{q_i}$. Assume that $\frac{d^b s_0}{dt^b}$ can be decomposed as a sum of the above form, so $\frac{d^b s_0}{dt^b} = \sum_{i=1}^z s_{b_i}$. Then

$$\frac{d^{b+1} s_0}{dt^{b+1}} = \frac{d}{dt} \left(\frac{d^b s_0}{dt^b} \right) = \sum_{i=1}^z \frac{d}{dt} (s_{b_i}) = \sum_{i=1}^z \sum_{j=1}^3 s_{i_j} \quad (57)$$

This shows by induction that $\frac{d^b s_0}{dt^b}$ can always be expressed as a sum of s_i .

Now we will compare the coefficients of s_1, s_2, s_{q_i} . Notice that for the coefficients, s_1 has an extra partial derivative with respect to t compared to s_0 , s_2 has an extra partial derivative with respect to u compared to s_0 , and s_{q_i} remains the same as s_0 .

Now we will compare the order and width of the monomials of s_1, s_2, s_3 . $Q(s_1) = Q(h)$, $W(s_1) = W(h)$, so the monomial term remains the same as s_0 . $Q(s_2) = Q(h) + 1$, $W(s_1) = W(h) + 1$, so both the order and width of the monomial term is one larger than s_0 . $Q(s_{q_i}) = Q(h) + 1$, $W(s_1) = W(h)$, so the order of the monomial term is one larger than s_0 .

Recall that

$$\phi^{t^{(n)}} = \frac{d^n}{dt^n} (\phi(t, u)) - \frac{d^n}{dt^n} (\tau(t, u) u_t) + \tau(t, u) \frac{\partial u_{t^{(n)}}}{\partial t} \quad (58)$$

and consider the first term, which takes derivatives of $\phi(t, u) \cdot 1$, which is of the form of s_i . So from above, $\frac{d^n}{dt^n} (\phi(t, u))$ can be expressed as a sum of functions of the form of s_i . Also, notice that $\phi(t, u)$ has both order and width zero, so the terms in $\frac{d^n}{dt^n} (\phi(t, u))$ have at most order/width n , since each derivative either keeps the order/width unchanged, or increases it by one.

Now consider the term in $\phi^{t^{(n)}}$ which has monomial term $m = \prod_{i=1}^p u_{t^{(k_i)}}$ (or $m = 1$). Let $y = Q(m) = \sum_{i=1}^p k_i$. Also, $p = W(m)$. Then from above, it necessarily follows that the corresponding coefficient is $c_1(m) \phi_{t^{(n-y)}u^{(p)}}$, where $c_1(m)$ is some non-negative integer. Clearly, $n - y$ must be a non-negative integer, so there is a restriction $y \leq n$. In other words, the order of the monomial term must not be greater than n . So if this restriction is met, then $c_1(m) \geq 1$, since it is always possible to get the corresponding monomial term in $\phi^{t^{(n)}}$ by following the relevant terms and their relevant derivatives.

The above holds the exact same for $g(j, k) = g = c \tau_{t^{(j)}u^{(k)}}$. Now consider the second term of Eq. 58, which takes derivatives of $\tau(t, u) \cdot u_t$, which is of the form of s_i . Again, $\frac{d^n}{dt^n} (\tau(t, u) u_t)$ can be expressed as a sum of functions of the form of s_i . Also, notice that $\tau(t, u) u_t$ has both order one and width zero, so the terms in $\frac{d^n}{dt^n} (\phi(t, u))$ have at most order $n + 1$ and width n , since each derivative

either keeps the order/width unchanged, or increases it by one.

Again consider the term in $\phi^{t^{(n)}}$ which has monomial term $m = \prod_{i=1}^p u_{t^{(k_i)}}$ (or $m = 1$). Similar to the ϕ terms, it necessarily follows that the corresponding coefficient is $c_2(m)\tau_{t^{(n-y+1)}u^{(p-1)}}$, where $c_2(m)$ is some non-negative integer, $y = Q(m) = \sum_{i=1}^p k_i$, and $p = W(m)$. Clearly, both $n - y + 1$ and $p - 1$ must be non-negative integers, so there is a restriction $y \leq n + 1$ and $p \geq 1$. In other words, the order of the monomial term must not be greater than $n + 1$, and the monomial must not be the constant one. So if this restriction is met, then $c_2(m) \geq 1$, since it is always possible to get the corresponding monomial term in $\phi^{t^{(n)}}$ by following the relevant terms and their relevant derivatives.

Finally, for completeness, consider the third term $\tau(t, u) \frac{\partial u_{t^{(n)}}}{\partial t}$. This term has order $n + 1$, and width 1.

Let us now investigate specific monomials in $\phi^{t^{(n)}}$.

First consider the coefficient for the $u_{tt}u_{t^{(n-1)}}$ term, which has order $n + 1$ and width 2. Since the order is larger than n , there are no corresponding terms with coefficients involving ϕ , or any of its partial derivatives. The third term does not match the width of two. So, this means that the only coefficients with this monomial come from the second term. The relevant coefficient will be $c_2(m)\tau_{t^{(n-(n+1)+1)}u^{(2-1)}} = c_2(m)\tau_{t^{(0)}u^{(1)}}$. So looking at the coefficients of terms with monomials $u_{tt}u_{t^{(n-1)}}$ in the constraint (Eq. 38) gives $-k_1\tau_u = 0$, which gives e in Table 2.

Next consider the coefficient for the $u_t u_{t^{(n-1)}}$ term, which has order n and width 2. This means that the only coefficients with this monomial come from the first and second term. The relevant coefficients will be $c_1(m)\phi_{t^{(n-n)}u^{(2)}} = c_2(m)\phi_{t^{(0)}u^{(2)}}$ and $c_2(m)\tau_{t^{(n-(n+1)+1)}u^{(2-1)}} = c_2(m)\tau_{t^{(1)}u^{(1)}}$. So looking at the coefficients of terms with monomials $u_{tt}u_{t^{(n-1)}}$ in the constraint (Eq. 38) gives $k_4\phi_{uu} - k_5\tau_{tu} = 0$, which gives d in Table 2.

Next, consider the coefficient for the $u_{t^{(n-1)}}$ term, which has order $n - 1$ and width 1. Similarly, the relevant coefficients will be $c_1(m)\phi_{t^{(n-(n-1))}u^{(1)}} = c_2(m)\phi_{t^{(1)}u^{(1)}}$ and $c_2(m)\tau_{t^{(n-(n-1)+1)}u^{(1-1)}} = c_2(m)\tau_{t^{(2)}u^{(0)}}$. So looking at the coefficients of terms with monomials $u_{t^{(n-1)}}$ in the constraint (Eq. 38) gives $n\phi_{uu} - \frac{n(n-1)}{2}\tau_{tu} = 0$, which gives c in Table 2 (Refer to appendix for proof of the coefficients, for the following terms below as well).

Next, consider the coefficient for the $u_{t^{(n-2)}}$ term, which has order $n - 2$ and width 1. Similarly, the relevant coefficients will be $c_1(m)\phi_{t^{(n-(n-2))}u^{(1)}} = c_2(m)\phi_{t^{(2)}u^{(1)}}$ and $c_2(m)\tau_{t^{(n-(n-2)+1)}u^{(1-1)}} = c_2(m)\tau_{t^{(3)}u^{(0)}}$. So looking at the coefficients of terms with monomials $u_{t^{(n-2)}}$ in the constraint (Eq. 38) gives $\frac{(n-1)n}{2}\phi_{ttu} - \frac{(n-2)(n-1)n}{6}\tau_{ttt} = 0$, which gives b in Table 2.

Finally, consider the coefficient for the constant monomial term, which has order 0 and width 0. The only coefficients with this monomial come from the first term. The relevant coefficients will be $c_1(m)\phi_{t^{(n-0)}u^{(0)}} = c_2(m)\phi_{t^{(n)}u^{(0)}}$. However, we must also consider the coefficient for the $u_{t^{(n)}}$ term, which has order n and width 1, since this turns into a constant monomial when Eq. 35 is substituted in. The relevant coefficients will be $c_1(m)\phi_{t^{(n-n)}u^{(1)}}\alpha u^3 = c_2(m)\phi_{t^{(0)}u^{(1)}}\alpha u^3$ and $c_2(m)\tau_{t^{(n-(n+1)}u^{(1-1)}}\alpha u^3 = c_2(m)\tau_{t^{(1)}u^{(0)}}\alpha u^3$. So looking at the coefficients of terms with constant

monomial in the constraint (Eq. 38) gives $\phi_{t^{(n)}} + \phi_u \alpha u^3 - n\tau \alpha u^3 = 3\alpha \phi u^3$, which gives a) in Table 2.

5. DISCUSSION

As seen above, for a general n there are exactly two symmetries, ∂_t which is known as time symmetry, and $t\partial_t - \frac{n}{2}u\partial_u$ which is known as scaling symmetry. Physically, time symmetry refers to how the soliton can be arbitrarily translated in time. This is an obvious property of the soliton, which has been known from the construction and physics of the system, however it is reassuring to see this symmetry emerging through the Lie symmetries method. Physically, scaling symmetry refers to a scaling in the time direction coming with a corresponding scaling in amplitude (specifically, making the soliton narrow in time by some factor corresponds to making the soliton taller by the same factor raised to the power $\frac{n}{2}$). This scaling is much less intuitive, but in fact has been seen experimentally for orders up to around ten. Again, it is reassuring to see this emerge from the Lie symmetries method, and we have extended beyond experiment to show that this pattern follows for any arbitrary order. It is important to emphasise that we have shown these are the *only* two symmetries that exist.

However, the results obtained are for the particular constraint where $\mu = 0$. This makes the differential equations significantly simpler, as it completely removes one of the terms. However, $\mu \neq 0$ is important, as physically, μ is a proxy for the strength of the nonlinear effect. Furthermore, mathematically it would dictate exponential decay in the tails of the solitons. This is not what we obtain, evident in the group invariance solutions, which decayed as $t^{-\frac{n}{2}}$, which is clearly not exponential. Nonetheless, we believe that these two symmetries are the only symmetries even for $\mu \neq 0$.

Additionally, these results only cover the pure solitons. Further symmetries may appear for mixed solitons, even in the simplest mixed case, which would be the combination of second and fourth order dispersion. Since the mixed cases are completely different form to the pure cases, we have no reason to suspect that only two symmetries exist.

6. CONCLUSION

For the pure, quartic soliton, we have explicitly calculated the two symmetries of the soliton, relating to time and scaling symmetry (for the particular constraint with $\mu = 0$). We have then successfully generalised this result for the pure n th order soliton, showing that there exists *only* two symmetries (again with the same constraint). This matches experiment, as these symmetries have been physically been observed. This shows that in this constraint, there are no other solutions that remain to be found.

For future work, the Lie symmetry method should be applied to pure solitons for $\mu \neq 0$. Additionally, mixed solitons should also be investigated with the Lie symmetry method (even with $\mu = 0$, as it will give insight to what symmetries might occur for $\mu \neq 0$). Overall, there is still much to discover about the symmetries and properties of solitons governed by the Generalised, Nonlinear Schrödinger Equation.

REFERENCES

- [1] Govind P. Agrawal. Chapter 5 - optical solitons. In Govind Agrawal, editor, *Nonlinear Fiber Optics (Fifth Edition)*, Optics and Photonics, pages 129–191. Academic Press, Boston, fifth edition edition, 2013.

- [2] A. Blanco-Redondo, C. Martijn de Sterke, J.E. Sipe, Thomas F. Krauss, Benjamin J. Eggleton Eggleton, and Chad Husko. Pure-quartic solitons. *Nature Communications*, 7:10427, 2016.
- [3] S. Burger, K. Bongs, S. Dettmer, W. Ertmer, K. Sengstock, A. Sanpera, G. V. Shlyapnikov, and M. Lewenstein. Dark solitons in Bose-Einstein condensates. *Phys. Rev. Lett.*, 83:5198–5201, Dec 1999.
- [4] C. Martijn de Sterke, Antoine F. J. Runge, Darren D. Hudson, and Andrea Blanco-Redondo. Pure-quartic solitons and their generalizations—Theory and experiments. *APL Photonics*, 6(9):091101, 09 2021.
- [5] R. K. Dodd, H. C. Morris, J. C. Eilbeck, and J. D. Gibbon. *Solitons and nonlinear wave equations*. Academic Press, New York, 1982.
- [6] Akira Hasegawa and Yoshiaki Kodama. *Solitons in Optical Communications*. Oxford University Press, 1995.
- [7] M. Kono and Miloš M. Škorić. In *Nonlinear physics of plasmas*, 2010.
- [8] Peter J. Olver. *Applications of Lie groups to differential equations*, volume 107 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, 1986.
- [9] Antoine F. J. Runge, Darren D. Hudson, Kevin K. K. Tam, C. Martijn de Sterke, and Andrea Blanco-Redondo. The pure-quartic soliton laser. *Nature Photonics*, 14(8):492–497, aug 2020.
- [10] Antoine F. J. Runge, Y. Long Qiang, Tristram J. Alexander, M. Z. Rafat, Darren D. Hudson, Andrea Blanco-Redondo, and C. Martijn de Sterke. Infinite hierarchy of solitons: Interaction of kerr nonlinearity with even orders of dispersion. *Phys. Rev. Res.*, 3:013166, Feb 2021.
- [11] Kevin K. K. Tam, Tristram J. Alexander, Andrea Blanco-Redondo, and C. Martijn de Sterke. Stationary and dynamical properties of pure-quartic solitons. *Opt. Lett.*, 44(13):3306–3309, Jul 2019.

APPENDIX A. DEFINITIONS

Definition A.1 (one-parameter Lie Group). A one-parameter Lie group is a group G which carries the structure of a one-dimensional manifold such that the group operation

$$m : G \times G \rightarrow G, \quad m(g, h) = g \cdot h, \quad g, h \in G$$

and the inversion

$$i : G \rightarrow G, \quad i(g) = g^{-1}, \quad g \in G$$

are smooth maps between manifolds.

A very simple example of such a Lie group is $G = \mathbb{R}$ with the group operations as usual $(x, y) \rightarrow x + y$, and the inverse $x \rightarrow -x$. Both operations are smooth and so R is a one-parameter abelian Lie Group. This definition can be extended to r -parameter Lie groups where r is arbitrary, but that is beyond our use case.

Definition A.2 (one-parameter Local Lie Group). A one-parameter Local Lie Group consists of connected open sets $\mathcal{U}_0 \subset \mathcal{U} \subset \mathbb{R}$ containing the origin 0, and smooth maps

$$m : \mathcal{U} \times \mathcal{U} \rightarrow \mathbb{R}$$

defining the group operation and

$$i : \mathcal{U}_0 \rightarrow \mathcal{U}$$

defining the group inversion with the usual associativity, identity, and inverse properties of a group.

Definition A.3 (Local Group of Transformations). Let M be a smooth manifold, and G a local Lie group. A local group of transformations acting on M is a smooth map from $\mathcal{U} \subset G \times M$ onto M with $\{e\} \times M \subset \mathcal{U}$ an open subset where e is the identity element of G . The map is specified as

$$(g, p) \rightarrow g \cdot p, \quad g \in G, \quad p \in M,$$

and satisfies the conditions below:

- $e \cdot p = p$ for all $p \in M$,
- $g_1 \cdot (g_2 \cdot p) = (g_1 \cdot g_2) \cdot p$, for $g_1, g_2 \in G$ and $p \in M$,
- $g^{-1} \cdot (g \cdot p) = p$, for $g \in G$ and $p \in M$ provided that $g \cdot p$ is defined.

As a consequence of these properties, we see that each group transformation is a diffeomorphism where it is defined.

Definition A.4 (Vector Field on a Smooth Manifold). A vector field X on a smooth manifold M is derivation acting on the space of smooth functions $C^\infty(M)$, defined as

$$X : C^\infty(M) \rightarrow C^\infty(M)$$

satisfying the rule

$$X(fg) = X(f)g + fX(g) \quad \text{for all } f, g \in C^\infty(M).$$

A vector field is often expressed in local coordinates $(x^1, x^2, \dots, x^n)$ on an n -dimensional manifold. In that case

$$X = \sum_{i=1}^n \xi^i(x) \frac{\partial}{\partial x^i} \quad (59)$$

where $\xi^i : M \rightarrow \mathbb{R}$ are smooth functions representing the components of the vector field in the chosen coordinate system.

APPENDIX B.

Proof. This is a proof of concept.

Assume that $n\phi_{tu}u_{t(n-1)}$ is a term in the expansion of $\phi^{t(n)}$. Then since $\phi^{t(n+1)} = \frac{d}{dt} \left(\phi^{t(n)} \right) - (\tau_t + \tau_u u_t)u_{t(n)}$, the term with the monomial $u_{t(n)}$ must come from either the $u_{t(n-1)}$ or $u_{t(n)}$ monomial from $\phi^{t(n)}$. Since $u_{t(n)}$ has order n and width 1, it follows that the corresponding coefficient is $1 \cdot \phi_u$. So differentiating and getting the relevant monomial gives $1 \cdot \phi_{tu}u_{t(n)}$. Now also differentiating $n\phi_{tu}u_{t(n-1)}$ to get the relevant term $n\phi_{tut}u_{t(n)}$, and collecting like terms gives a total of $(n+1)\phi_{tut}u_{t(n)}$ in the expansion of $\phi^{t(n+1)}$. So the coefficient of $u_{t(n)}$ in the expansion of $\phi^{t(n)}$ is $n\phi_{tu}$ by induction (base case of $n = 2$ omitted, as it is straightforward).

The idea is the same for any other coefficient where a specific scalar coefficient is required. The proof is done by induction, using the recursive formula for $\phi^{t(n-1)}$. For example, for the monomial term $u_{t(n-2)}$, the terms from $\phi^{t(n-1)}$ which could turn into the correct monomial term is either $u_{t(n-2)}$ or $u_{t(n-3)}$. Then, the relevant coefficients of ϕ, τ is considered, along with the order and width of the $u_{t(n-2)}$. The different ways to achieve the $u_{t(n-2)}$ term is then added to get the correct coefficient. $\square$